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(54) **SELF-ALIGNING TACTICAL BELT**

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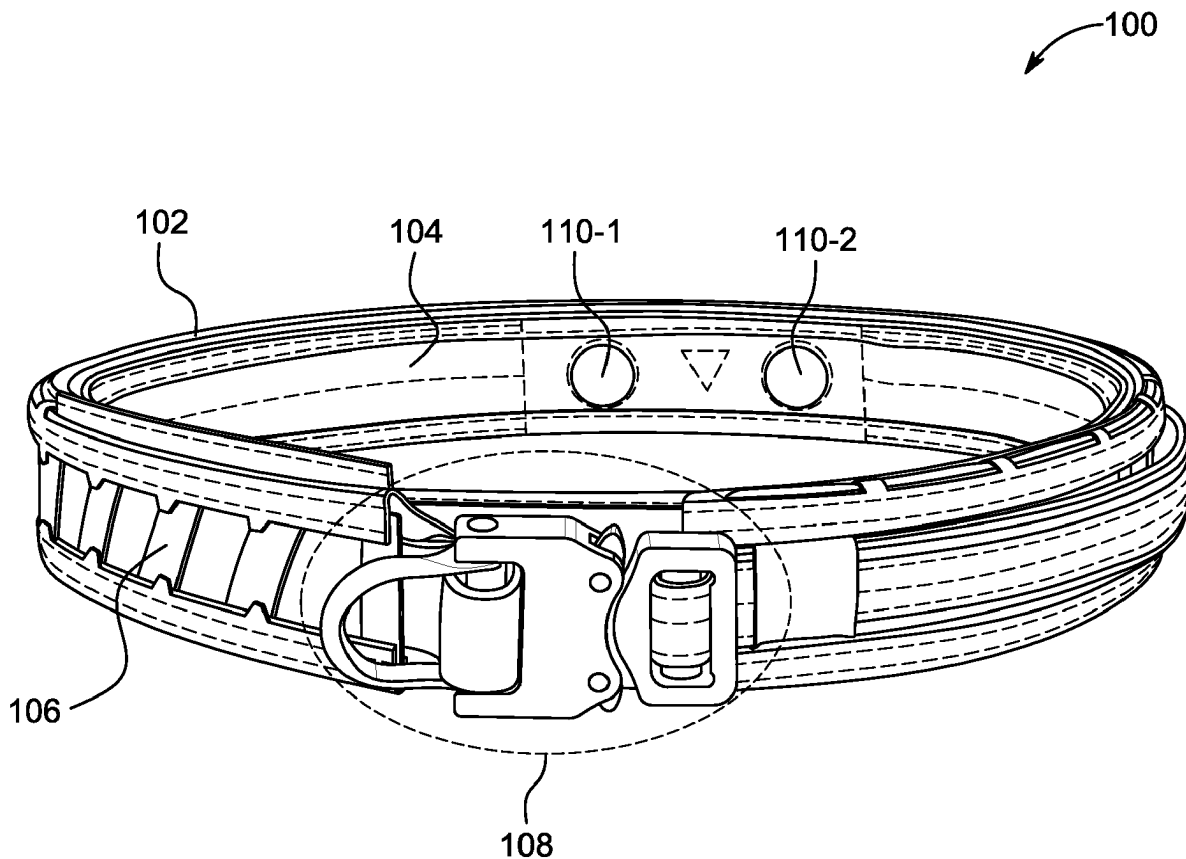
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(57) **ABSTRACT**

In one embodiment, a self-aligning tactical belt may include an inner belt portion having, on a posterior area of the inner belt portion, a first inner belt magnetic component and a second inner belt magnetic component coupled thereto and an outer belt portion having, on a posterior area of the outer belt portion, a first outer belt magnetic component and a second outer belt magnetic component coupled thereto. The first inner belt magnetic component and the second inner belt magnetic component may be configured to couple to the first outer belt magnetic component and the second outer belt magnetic component, respectively, to facilitate alignment of the inner belt portion and the outer belt portion.



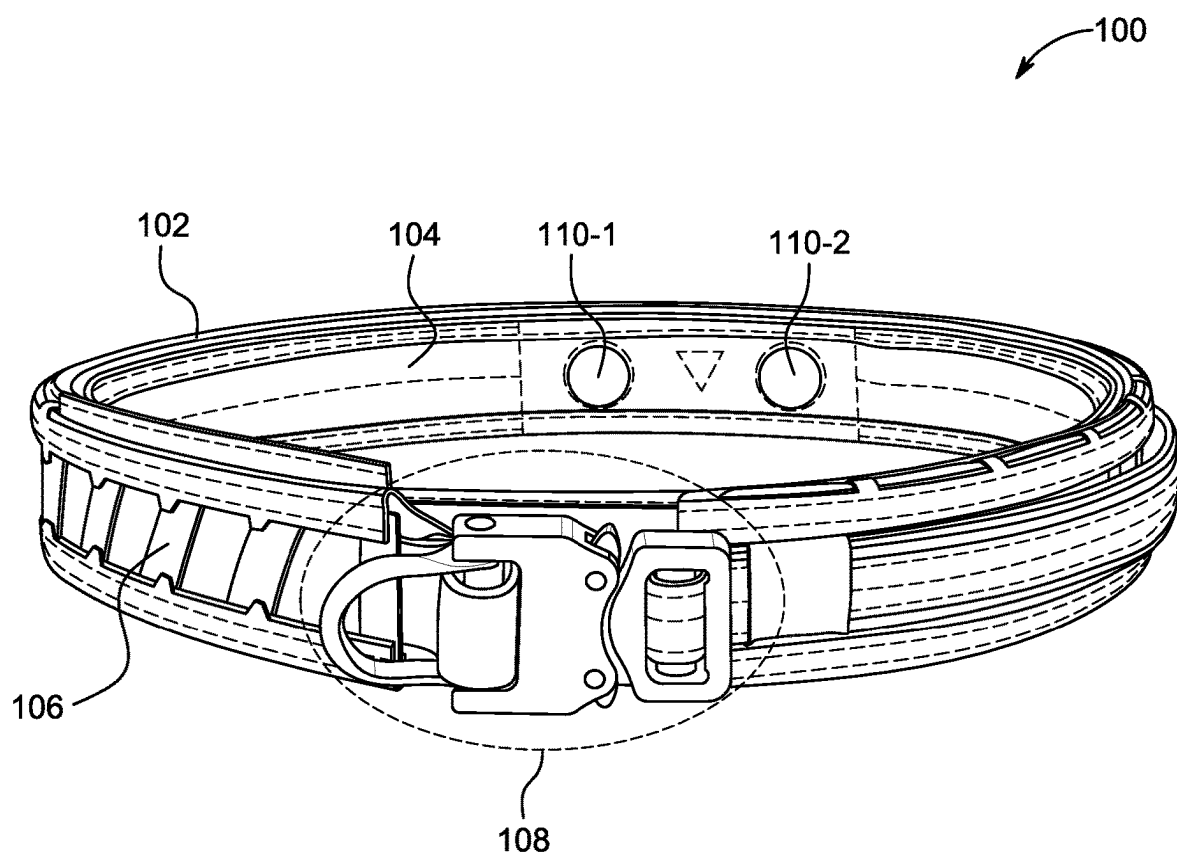


FIG. 1

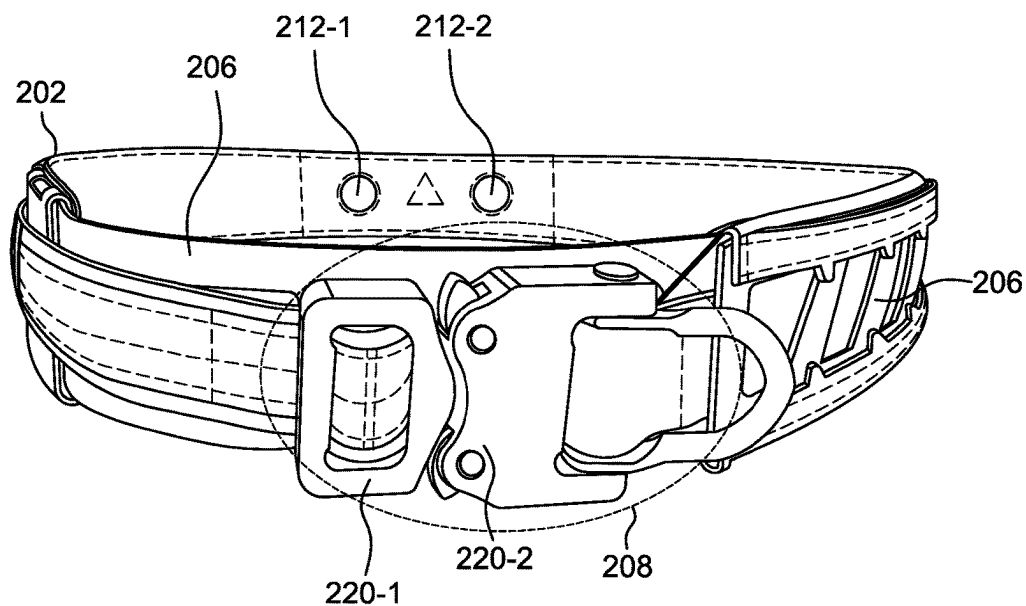


FIG. 2A

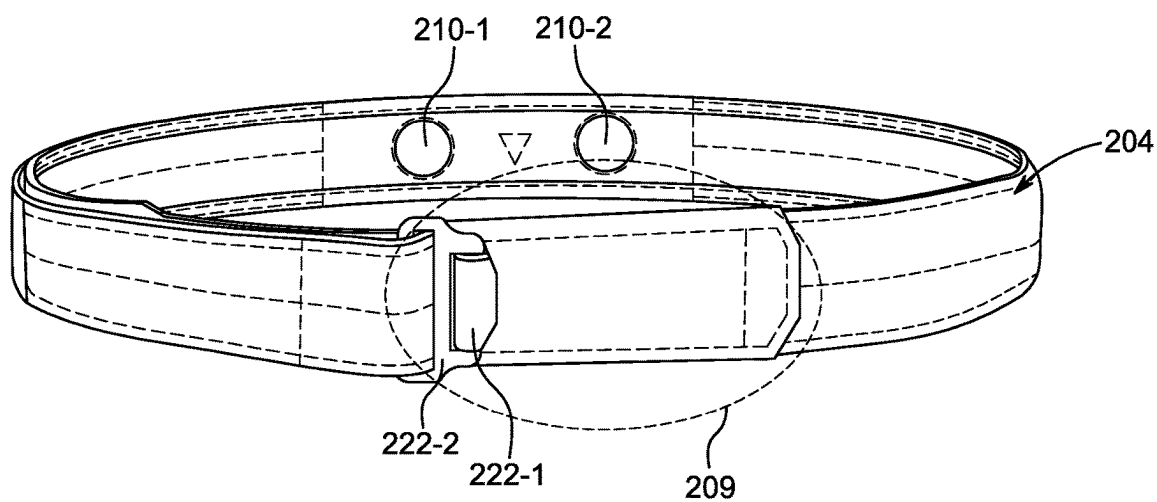


FIG. 2B

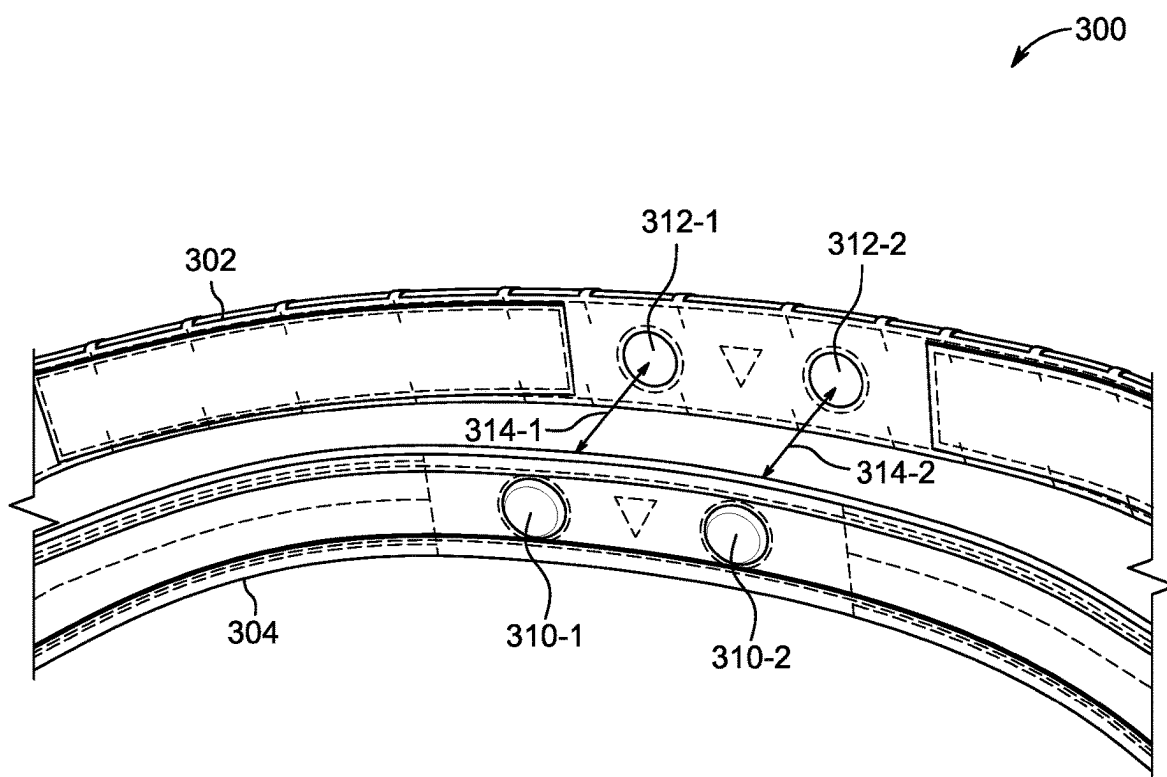


FIG. 3

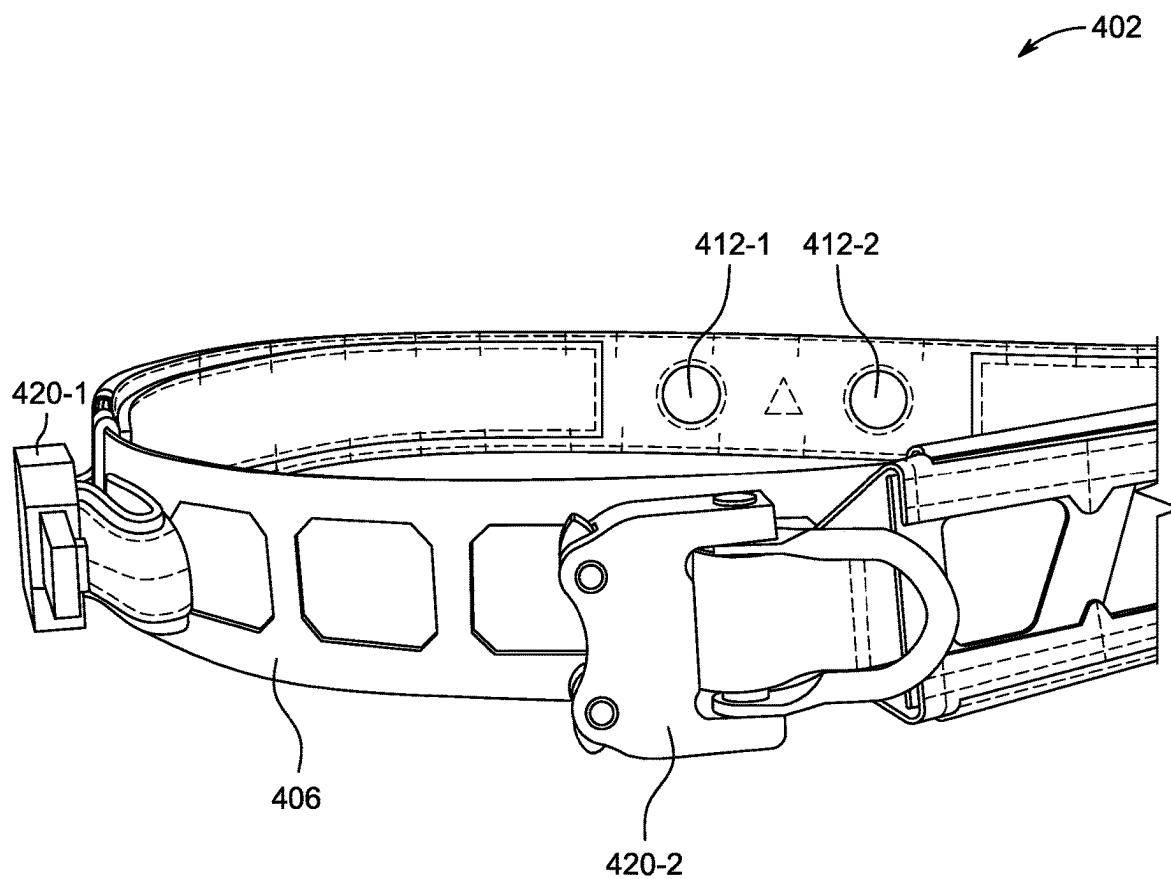


FIG. 4A

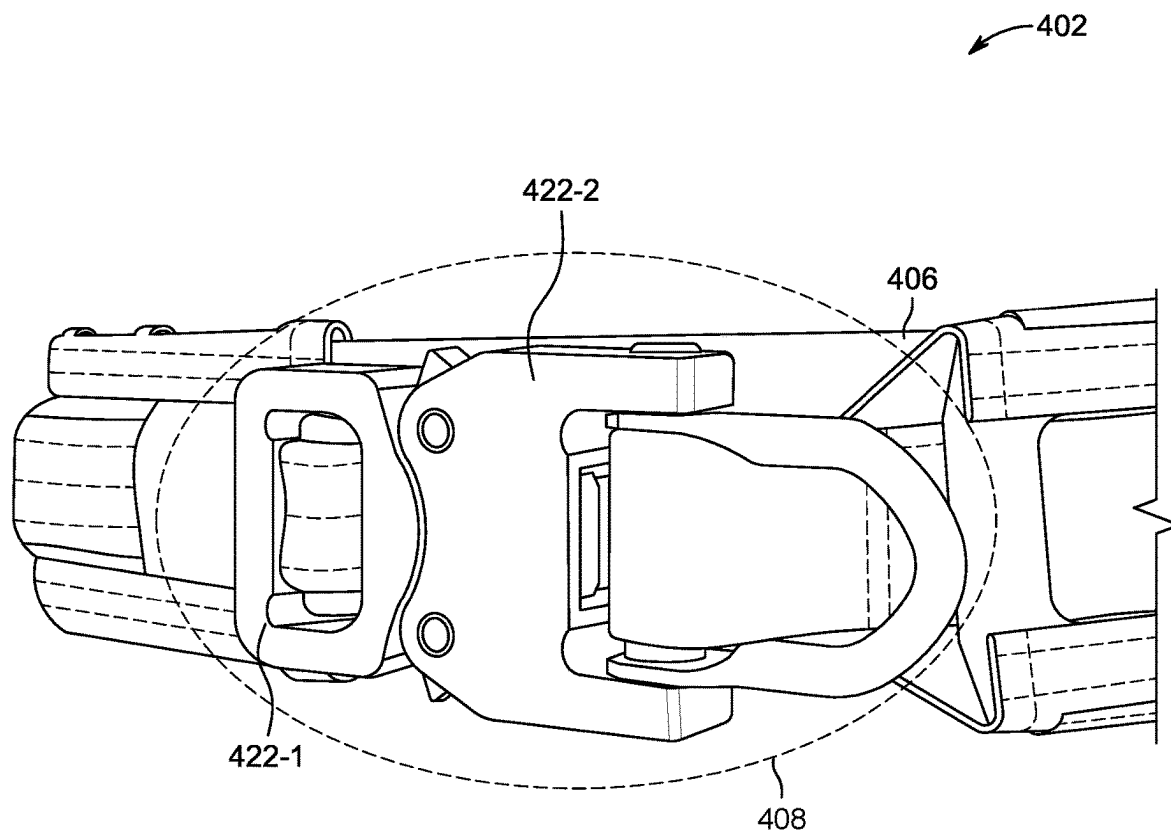


FIG. 4B

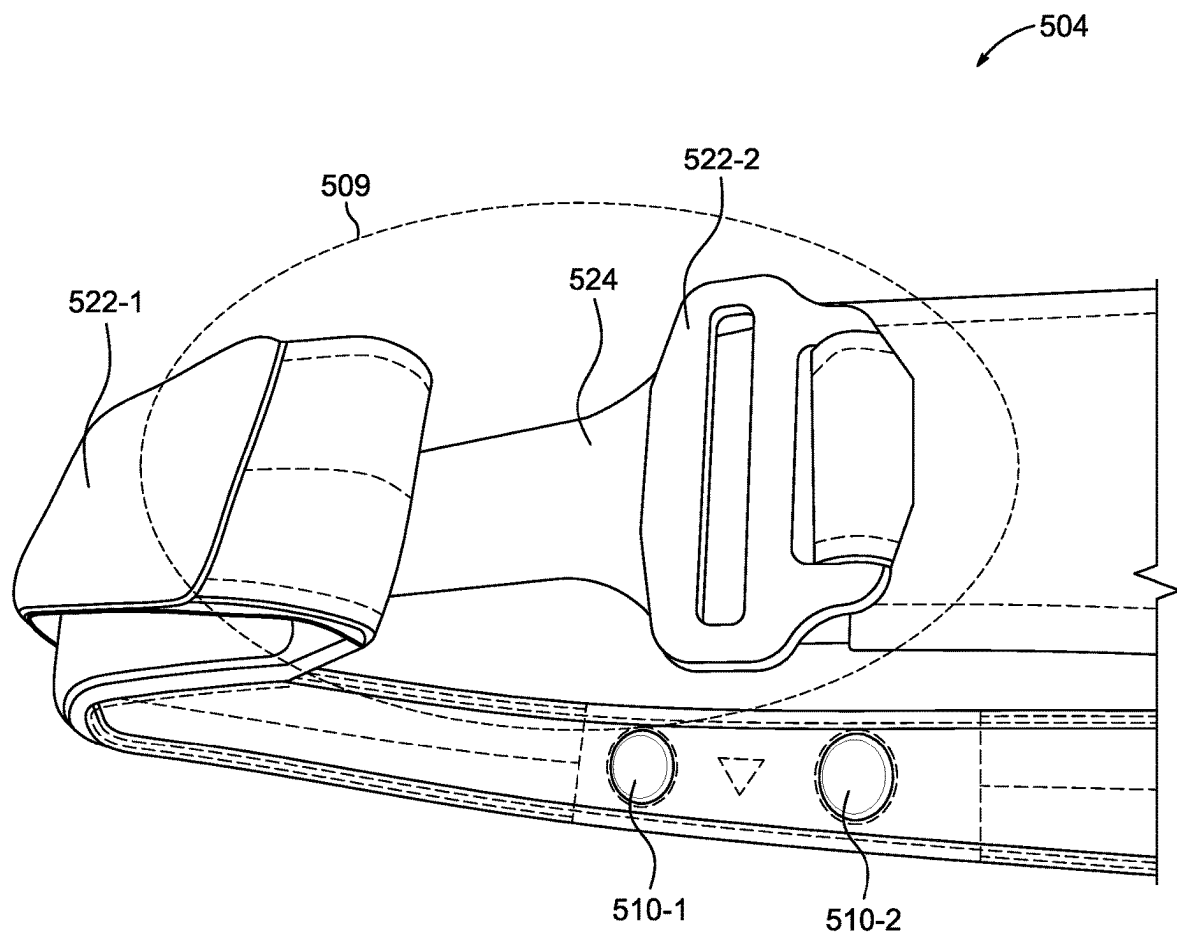


FIG. 5

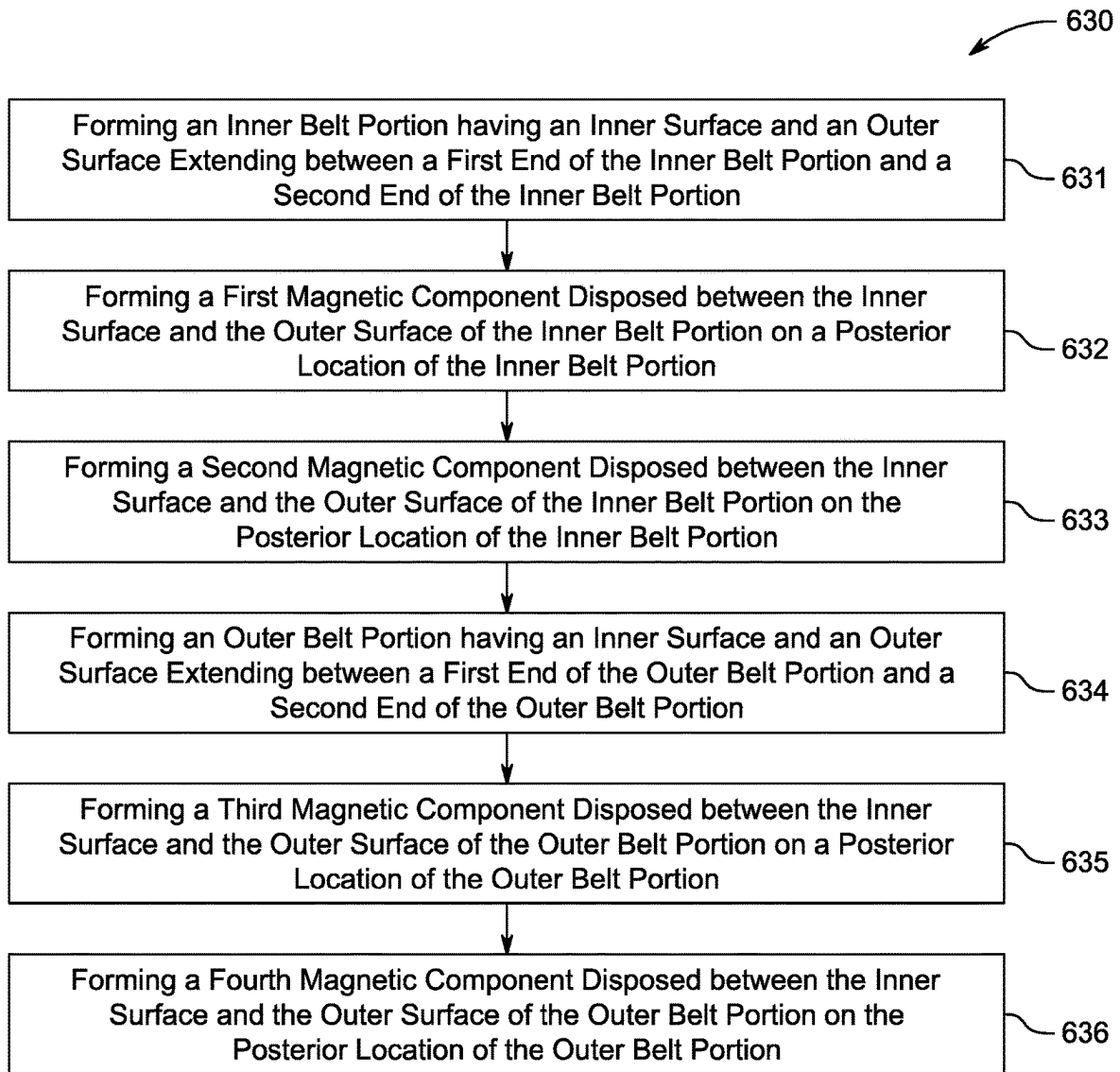


FIG. 6

SELF-ALIGNING TACTICAL BELT

TECHNICAL FIELD

[0001] The present invention relates to the field of equipment bearing garments such as tactical belts for sporting, military, law enforcement, and tactical applications. More specifically, the invention pertains to a self-aligning tactical belt that offers enhanced ease of equipping, comfort, mobility, load leveling, and customization options for users.

BACKGROUND

[0002] In tactical situations, personnel, such as military or law enforcement personnel, often depend on personal mobility to transport mission specific equipment. In general, at least a portion of this mission specific equipment is carried using a belt, such as a “tactical belt,” mounted on the waist of an individual. The type of equipment may vary significantly depending on the specific mission or tactical situation.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The embodiments herein may be better understood by referring to the following description in conjunction with the accompanying drawings in which like reference numerals indicate identically or functionally similar elements, of which:

[0004] FIG. 1 illustrates a perspective view of an example self-aligning tactical belt;

[0005] FIG. 2A illustrates an example of an outer belt portion of a self-aligning tactical belt;

[0006] FIG. 2B illustrates an example of an inner belt portion of a self-aligning tactical belt;

[0007] FIG. 3 illustrates an example of a self-aligning tactical belt having an inner belt portion and an outer belt portion;

[0008] FIG. 4A illustrates an example of the outer belt portion of the self-aligning tactical belt when the buckle is in a disengaged position;

[0009] FIG. 4B illustrates an example of the outer belt portion of the self-aligning tactical belt when the buckle is in an engaged position;

[0010] FIG. 5 illustrates an example of the inner belt portion of the self-aligning tactical belt when the buckle is in a disengaged position; and

[0011] FIG. 6 illustrates an example of a method for fabricating the self-aligning tactical belt.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Overview

[0012] According to one or more embodiments of the disclosure, a self-aligning tactical belt includes an inner belt portion having, on a posterior area of the inner belt portion, a first inner belt magnetic component and a second inner belt magnetic component coupled thereto and an outer belt portion having, on a posterior area of the outer belt portion, a first outer belt magnetic component and a second outer belt magnetic component coupled thereto. In such embodiments, the first inner belt magnetic component and the second inner belt magnetic component are configured to couple to the first outer belt magnetic component and the second outer belt magnetic component, respectively, to facilitate alignment of the inner belt portion and the outer belt portion.

[0013] According to other embodiments of the disclosure, a self-aligning tactical belt includes an inner belt portion having an inner surface and an outer surface extending between a first end of the inner belt portion and a second end of the inner belt portion. The inner belt portion includes a first magnetic component disposed between the inner surface and the outer surface of the inner belt portion and a second magnetic component disposed between the inner surface and the outer surface of the inner belt portion. As shown in more detail below, the first magnetic component and the second magnetic component are disposed on a posterior location of the inner belt portion.

[0014] In addition, the self-aligning tactical belt includes an outer belt portion having an inner surface and an outer surface extending between a first end of the outer belt portion and a second end of the outer belt portion. The outer belt portion includes a third magnetic component disposed between the inner surface and the outer surface of the outer belt portion and a fourth magnetic component disposed between the inner surface and the outer surface of the outer belt portion. As shown in more detail below, the third magnetic component and the fourth magnetic component are disposed on a posterior location of the outer belt portion.

[0015] The first magnetic component disposed between the inner surface and the outer surface of the inner belt portion is couplable to the third magnetic component disposed between the inner surface and the outer surface of the outer belt portion, and the second magnetic component disposed between the inner surface and the outer surface of the inner belt portion is couplable to the fourth magnetic component disposed between the inner surface and the outer surface of the outer belt portion.

[0016] According to yet other embodiments of the disclosure, a method for fabricating a self-aligning tactical belt includes forming an inner belt portion having an inner surface and an outer surface extending between a first end of the inner belt portion and a second end of the inner belt portion; forming a first magnetic component disposed between the inner surface and the outer surface of the inner belt portion on a posterior location of the inner belt portion; forming a second magnetic component disposed between the inner surface and the outer surface of the inner belt portion on the posterior location of the inner belt portion; forming an outer belt portion having an inner surface and an outer surface extending between a first end of the outer belt portion and a second end of the outer belt portion; forming a third magnetic component disposed between the inner surface and the outer surface of the outer belt portion on a posterior location of the outer belt portion; and forming a fourth magnetic component disposed between the inner surface and the outer surface of the outer belt portion on the posterior location of the outer belt portion.

[0017] Other embodiments are described below, and this overview is not meant to limit the scope of the present disclosure.

DESCRIPTION

[0018] Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the application is not limited to the details or methodologies set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

[0019] As noted above, in tactical situations, personnel, such as military or law enforcement personnel, often depend on personal mobility to transport mission specific equipment. To this end, the use of tactical belts, which allow for mission specific equipment to be mounted thereto, is quite common. Accordingly, there are a wide range of tactical belts available on the market today. Some tactical belts are formed of two separable belt portions, which can be referred to as an “inner belt portion” and an “outer belt portion,” herein. It is noted that this terminology is used for ease of describing the embodiments disclosed herein and that an “inner belt portion” may be, in the alternative, referred to as a “lower portion,” while an “outer belt portion” may be, in the alternative referred to as an “upper portion” in some parlance.

[0020] As discussed in more detail below, the outer belt portion can include one or more rows of webbing adapted to carry mission specific equipment that may be interchanged based on the needs of the mission. For example, the one or more rows of webbing coupled to the outer belt portion may be Modular Lightweight Load-carrying equipment (MOLLE) webbing, or other suitable webbing(s) that can allow for mission specific equipment or “tactical accessories” (e.g., holsters, pouches, medical kits, radios, wires, drink bladders, tools, etc.) to be attached thereto.

[0021] In accordance with the disclosure, the inner belt portion is generally more light weight and can be comfortably worn for an extended period of time, while the outer belt portion may be more rugged and may generally be worn for a shorter period of time than the inner belt portion. More specifically, the inner belt portion described herein can be worn as a “regular” or “daily” belt (e.g., an everyday conceal carry belt) that is comfortable enough that a user may wear the inner belt portion for an extended period of time and/or during normal day-to-day activities, while the outer belt portion may be equipped in certain situations and/or circumstances. As a result, tactical belts that are formed of two separable belt portions can allow the user to remove the outer belt portion when not necessary for improved comfort, reduced weight, etc. while still allowing for the inner belt portion to be worn.

[0022] When a tactical situation presents itself, the outer belt portion should be capable of being quickly and easily coupled to the inner belt portion. However, the outer belt portion of current two-piece tactical belts is often difficult to quickly and easily align with the inner belt portion. For example, when attempting to couple the outer belt portion to the inner belt portion, it can take several tries to ensure that the buckles on the anterior sections of the inner belt portion and outer belt portion align so that the buckle may be engaged. This can lead to wasted time in having to disattach and reattach the outer belt portion to the inner belt portion to properly align the two belt portions to engage the buckle, which can lead to unnecessary delays in preparing for tactical situations, particularly those tactical situations in which time is of the essence.

[0023] In contrast, the self-aligning tactical belt described herein introduces magnetic components to a posterior area of the outer belt portion and magnetic components to a posterior area of the inner belt portion. These magnetic components allow for the two belt portions to self-align when the outer belt portion is being coupled to the inner belt portion, thereby mitigating the need to disattach and reattach the outer belt portion to the inner belt portion to properly

align the two belt portions to engage the buckle. This in turn reduces the time (and frustration) associated with coupling the outer belt portion to the inner belt portion that often plague current designs therefore leading to an enhanced user experience when compared to previous approaches.

[0024] FIG. 1 illustrates a perspective view of an example self-aligning tactical belt 100. As shown in FIG. 1, the self-aligning tactical belt 100 includes an outer belt portion 102 and an inner belt portion 104. In this view, the outer belt portion 102 and the inner belt portion 104 are coupled to one another. Although not explicitly shown so as to not obfuscate the drawing layout, the outer belt portion 102 and the inner belt portion 104 may be coupled to one another using a hook-and-loop fastening system (e.g., Velcro® or similar hook-and-loop fastening system). The self-aligning tactical belt 100, the outer belt portion 102, and/or the inner belt portion 104 can be any type of tactical belt such as, but not limited to: a load-bearing tactical belt, a conceal-carry tactical belt, etc. and can be made of various materials or combinations of materials such as, but not limited to: nylon, leather, composite materials, etc.

[0025] It will be appreciated that, for such a hook-and-loop fastening system to effectively couple the outer belt portion 102 and the inner belt portion 104, the hook-and-loop fastening system may be disposed on an interior area of the outer belt portion 102 and an exterior area of the inner belt portion 104. In some embodiments, the hook-and-loop fastening system may extend substantially along the exterior area of the inner belt portion 104 and substantially along the interior area of the outer belt portion. Embodiments are not so limited, however, and in some embodiments, the hook-and-loop fastening system may be provided in certain areas along the exterior area of the inner belt portion 104 and in certain areas along the interior area of the outer belt portion.

[0026] The outer belt portion 102 includes a skeletonized reinforcement band 106. The skeletonized reinforcement band 106 can be stiff or semi-rigid and can be constructed of various composite materials, such as plastics or other semi-rigid materials. In some embodiments, the skeletonized reinforcement band 106 can provide extra stability to the outer belt portion 102 and/or can provide enhanced rigidity to the outer belt portion to facilitate enhanced durability, strength, fit, and/or comfort for the user of the outer belt portion 102. In some embodiments, the skeletonized reinforcement band 106 can interface with (e.g., couple to) an attachment mechanism such as hook and loop fasteners, MOLLE webbing, Pouch Attachment Ladder System (PALS), buckles, hooks, snaps, quick release mechanisms, etc. that may be used to attach various tactical accessories (e.g., holsters, pouches, medical kits, radios, wires, drink bladders, tools, etc.). In some embodiments, the skeletonized reinforcement band 106 can have a spacing that corresponds to a MOLLE webbing spacing. Further, as described in more detail below in connection with FIG. 4A, the skeletonized reinforcement band 106 can extend continuously behind the buckle 108 such that the skeletonized reinforcement band 106 forms a ring around the entirety of the outer belt portion 102.

[0027] As shown in FIG. 1, the inner belt portion 104 includes a first magnetic component 110-1 and a second magnetic component 110-2. The first magnetic component 110-1 and the second magnetic component 110-2 are located on a posterior area of the inner belt portion 104 with respect to the view illustrated in FIG. 1. As will be discussed in more

detail below in connection with FIG. 3, the first magnetic component 110-1 and the second magnetic component 110-2 are located on an anterior area of the self-aligning tactical belt 100 with respect to the outer belt portion 102.

[0028] The first magnetic component 110-1 and the second magnetic component 110-2 may be disposed in between an inner surface and an outer surface of the inner belt portion 104. That is, in some embodiments, the first magnetic component 110-1 and the second magnetic component 110-2 can be sewn, stitched, or otherwise secured within a material that is used to form the inner belt portion 104 such that the first magnetic component 110-1 and the second magnetic component 110-2 are not exposed outside of the material that forms the inner belt portion 104. Embodiments are not so limited, however, and in other embodiments, the first magnetic component 110-1 and the second magnetic component 110-2 may be coupled to the inner belt portion 104 in a manner that leaves at least a portion of the first magnetic component 110-1 and/or the second magnetic component 110-2 exposed outside of the material that forms the inner belt portion 104.

[0029] FIG. 2A illustrates an example of an outer belt portion 202 of a self-aligning tactical belt (e.g., the self-aligning tactical belt 100 of FIG. 1). The outer belt portion 202 of FIG. 2A may be analogous to the outer belt portion 102 of FIG. 1. As shown in FIG. 2A, the outer belt portion 202 includes skeletonized reinforcement band 206, which may be analogous to the skeletonized reinforcement band 106 of FIG. 1. In addition, the outer belt portion 202 includes a buckle 208, which can be analogous to the buckle 108 of FIG. 1.

[0030] The buckle 208 can include a first buckle portion 220-1 and a second buckle portion 220-2 that can be engaged to secure the outer belt portion 202 and, hence, the self-aligning tactical belt, to a user of said belt. As will be appreciated, the first buckle portion 220-1 and the second buckle portion 220-2 can be disengaged to allow for removal of the self-aligning tactical belt and/or to allow for removal of the outer belt portion 202 from the inner belt portion 204 of FIG. 2B. In some embodiments, the buckle 208 may be a safety-rated buckle (e.g., the buckle 208 may be a load-bearing buckle).

[0031] As shown in FIG. 2A, the outer belt portion 202 includes a first magnetic component 212-1 and a second magnetic component 212-2. The first magnetic component 212-1 and the second magnetic component 212-2 are located on a posterior area of the outer belt portion 202 with respect to the view illustrated in FIG. 2A. It is noted that, particularly with respect to the claims and to provide clarity to the claims, that the first magnetic component 212-1 may be referred to as a “third magnetic component” and the second magnetic component 212-2 may be referred to as a “fourth magnetic component.” That is, in order to clearly distinguish the plurality of magnetic components described herein, the first magnetic component 210-1 of FIG. 2B and the second magnetic component 210-2 of FIG. 2B may be referred to, for brevity, as a first magnetic component and a second magnetic component, respectively while the first magnetic component 212-1 of FIG. 2A and the second magnetic component 212-2 of FIG. 2A may be referred to, for brevity, as a third magnetic component and a fourth magnetic component, respectively given the context of the disclosure.

[0032] The first magnetic component 212-1 (e.g., the third magnetic component) and the second magnetic component

212-2 (e.g., the fourth magnetic component) may be disposed in between an inner surface and an outer surface of the outer belt portion 202. That is, in some embodiments, the first magnetic component 212-1 and the second magnetic component 212-2 can be sewn, stitched, or otherwise secured within a material that is used to form the outer belt portion 202 such that the first magnetic component 212-1 and the second magnetic component 212-2 are not exposed outside of the material that forms the outer belt portion 202. Embodiments are not so limited, however, and in other embodiments, the first magnetic component 212-1 and the second magnetic component 212-2 may be coupled to the outer belt portion 202 in a manner that leaves at least a portion of the first magnetic component 212-1 and/or the second magnetic component 212-2 exposed outside of the material that forms the outer belt portion 202.

[0033] FIG. 2B illustrates an example of an inner belt portion 204 of a self-aligning tactical belt (e.g., the self-aligning tactical belt 100 of FIG. 1). The inner belt portion 204 of FIG. 2B may be analogous to the inner belt portion 104 of FIG. 1. The inner belt portion 204 includes a buckle 209. The buckle 209 can include a first buckle portion 222-1 and a second buckle portion 222-2 that can be engaged to secure the inner belt portion 204 to a user of said belt. In some embodiments, the second buckle portion 222-2 can allow for a portion of the first buckle portion 222-1 to pass through an opening in the second buckle portion 222-2. In such embodiments, the first buckle portion 222-1 can include a hook-and-loop fastener that can fold back on itself to secure the inner belt portion 204. As will be appreciated, the first buckle portion 222-1 and the second buckle portion 222-2 can be disengaged to allow for removal of the inner belt portion 204.

[0034] In addition, the inner belt portion 204 includes a first magnetic component 210-1 and a second magnetic component 210-2. The first magnetic component 210-1 and the second magnetic component 210-2, which may be analogous to the first magnetic component 110-1 and the second magnetic component 110-2 of FIG. 1, are located on a posterior area of the inner belt portion 204 with respect to the view illustrated in FIG. 2B.

[0035] FIG. 3 illustrates an example of a self-aligning tactical belt 300 (e.g., the self-aligning tactical belt 100 of FIG. 1) having an inner belt portion 304 and an outer belt portion 302. The inner belt portion 304 of FIG. 3 may be analogous to the inner belt portion 204 of FIG. 2B and/or the inner belt portion 104 of FIG. 1. Similarly, the outer belt portion 302 of FIG. 3 may be analogous to the outer belt portion 202 of FIG. 2A and/or the outer belt portion 102 of FIG. 1.

[0036] As shown in FIG. 3, the inner belt portion 304 includes a first magnetic component 310-1 and a second magnetic component 310-2 and the outer belt portion 302 includes a first magnetic component 312-1 and a second magnetic component 312-2. The first magnetic component 310-1 of the inner belt portion 304 can be analogous to the first magnetic component 210-1 of the inner belt portion 204 of FIG. 2B, while the second magnetic component 310-2 of the inner belt portion 304 can be analogous to the second magnetic component 210-2 of the inner belt portion 204 of FIG. 2B. Similarly, the first magnetic component 312-1 of the outer belt portion 302 can be analogous to the first magnetic component 212-1 of the outer belt portion 202 of FIG. 2A, while the second magnetic component 312-2 of the

outer belt portion **302** can be analogous to the second magnetic component **212-2** of the outer belt portion **202** of FIG. 2A.

[0037] In general, the magnetic components serve to align the outer belt portion **302** with the inner belt portion **304** to ensure that the outer belt portion **302** and the inner belt portion **304** match up for quick, easy, and accurate coupling of the outer belt portion **302** to the inner belt portion **304**. For example, the first magnetic component **310-1** of the inner belt portion **304** will be magnetically attracted to the first magnetic component **312-1** of the outer belt portion **302** (as indicated by the arrow **314-1**) while the second magnetic component **310-2** of the inner belt portion **304** will be magnetically attracted to the second magnetic component **312-2** of the outer belt portion **302** (as indicated by the arrow **314-2**). This, in turn, guarantees that the front regions of the outer belt portion **302** and the inner belt portion **304** will be aligned thereby allowing a user of the self-aligning tactical belt to engage the buckle (e.g., the buckle **208** of FIG. 2A) without having to readjust the outer belt portion **302** as can occur in previous approaches.

[0038] In some embodiments, the magnetic components of the outer belt portion **302** can have opposite polarities from the magnetic components of the inner belt portion **304**. For example, a north magnetic pole of the first magnetic component **312-1** of the outer belt portion **302** can be facing a south magnetic pole of the first magnetic component **310-1** of the inner belt portion **304**, or vice versa. Similarly, a north magnetic pole of the second magnetic component **312-2** of the outer belt portion **302** can be facing a south magnetic pole of the second magnetic component **310-2** of the inner belt portion **304**, or vice versa.

[0039] In other embodiments, a north magnetic pole of the first magnetic component **312-1** of the outer belt portion **302** can be facing a south magnetic pole of the first magnetic component **310-1** of the inner belt portion **304** while a south magnetic pole of the second magnetic component **312-2** of the outer belt portion **302** can be facing a north magnetic pole of the second magnetic component **310-2** of the inner belt portion **304**, or vice versa.

[0040] This can provide an enhanced attractive force between the magnetic components to assist in quickly aligning the outer belt portion **302** with the inner belt portion **304**. That is, because magnetic components having opposite polarities will attract and seek to couple to one another, the accurate coupling of the outer belt portion **302** to the inner belt portion **304** can be assured in embodiments that include these features.

[0041] FIG. 4A illustrates an example of the outer belt portion **402** of the self-aligning tactical belt (e.g., the self-aligning tactical belt **100** of FIG. 1) when the buckle (e.g., the buckle **108** of FIG. 1) is in a disengaged position. It will be appreciated that the outer belt portion **402** can be analogous to the outer belt portion **102/202/302** illustrated in FIG. 1, FIG. 2A, and FIG. 3, herein.

[0042] As shown in FIG. 4A, the outer belt portion **402** includes a first buckle portion **420-1** and a second buckle portion **420-2** that can be engaged to secure the outer belt portion **402** (as shown in FIG. 4B) and, hence, the self-aligning tactical belt, to a user of said belt. In addition, the outer belt portion **402** includes a first magnetic component **412-1** and a second magnetic component **412-2**, which can

be analogous to the first magnetic component **212-1/312-1** and the second magnetic component **212-2/312-2** of FIG. 2A and FIG. 3.

[0043] Further, the outer belt portion **402** includes a skeletonized reinforcement band **406**, which can be analogous to the skeletonized reinforcement band **106/206** of FIG. 1 and FIG. 2A. As mentioned above, when the outer belt portion **402** is in a state in which the buckle is disengaged, as shown in FIG. 4A, the skeletonized reinforcement band **406** may extend beyond a location where the material of the outer belt portion **402** ends one end of the outer belt portion **402** (e.g., through a void or gap created when the buckle is disengaged) past a location where the material of the outer belt portion **402** begins on an opposite end of the outer belt portion **402**. That is, and as will be apparent from the illustrations of FIG. 4A and FIG. 4B that, when the buckle is engaged, the skeletonized reinforcement band **406** will extend through the opposite side of the outer belt portion (e.g., behind the second buckle portion **420-2**). Stated alternatively, the skeletonized reinforcement band **406** can be provided in a “continuous” ring that surrounds the user when the buckle is engaged thereby providing the opportunity to couple a full field of webbing to the outer belt portion that can be expanded regardless of a physical size of the user.

[0044] These features can allow for the skeletonized reinforcement band **406** (and hence, webbing that may be coupled to the skeletonized reinforcement band) to be fully functional in locations (e.g., up to the buckle) where other approaches often experience shortcomings in this area. Further, in addition to the benefit of skeletonized reinforcement band **406** that is fully functional in these locations, by providing the skeletonized reinforcement band **406** that extends through the void or gap created when the buckle is disengaged, the rigidity and strength of the outer belt portion **402** can be enhanced making the outer belt portion **402** less likely to sag when encumbered with tactical accessories when compared to previous approaches.

[0045] FIG. 4B illustrates an example of the outer belt portion **402** of the self-aligning tactical belt (e.g., the self-aligning tactical belt **100** of FIG. 1) when the buckle **408** is in an engaged position. As mentioned above in connection with FIG. 4A, as shown in FIG. 4B, the skeletonized reinforcement band **406** extends behind both the first buckle portion **420-1** and the second buckle portion **420-2** when the buckle **408** is engaged. As mentioned above, this allows the skeletonized reinforcement band **406** to form a continuous ring that surrounds the user when the buckle is engaged thereby providing a full field of skeletonized reinforcement band **406** (and hence a full field of webbing, when such webbing is coupled to the skeletonized reinforcement band) regardless of the physical size of the user.

[0046] FIG. 5 illustrates an example of the inner belt portion **504** of the self-aligning tactical belt (e.g., the self-aligning tactical belt **100** of FIG. 1) when the buckle **509** is in a disengaged position. The inner belt portion **504** can be analogous to the inner belt portion **104/204/304** if FIG. 1, FIG. 2B, and FIG. 3 while the buckle **509** can be analogous to the buckle **209** of FIG. 2B, herein.

[0047] As shown in FIG. 5, the inner belt portion **504** includes a first buckle portion **522-1** and a second buckle portion **522-2**, which can be analogous to the first buckle portion **222-1** and second buckle portion **222-2** illustrated in FIG. 2B. As mentioned above, the first buckle portion **522-1** can be a hook-and-loop fastener that can have an end that is

configured to fit through an opening of the second buckle portion **522-2** and loop back around to secure the inner belt portion **504** to a user of the tactical belt. Although shown as a hook-and-loop fastener, the first buckle portion **522-1** may comprise different types of buckle portions that are configured to form the buckle **509**. Similarly, the second buckle portion **522-2** can be any type of buckle portion that interfaces with the first buckle portion **522-1** to secure the inner belt portion **504** to the user of the tactical belt.

[0048] As shown in FIG. 5, The inner belt portion **504** further includes a stiffening tongue **524**. The stiffening tongue **524** can be sewn or stitched into the materials that form the inner belt portion **504** and can extend behind both the first buckle portion **522-1** and the second buckle portion **522-2** when the buckle **509** is engaged. In some embodiments, the stiffening tongue **524** can provide a continuous connection across an anterior portion (with respect to the user of the tactical belt) of the inner belt portion **504** that can assist in keeping the inner belt portion **504** fastened while allowing for an additional degree of motion when the user of the tactical belt engages in motion.

[0049] For example, if the user bends forward with a firearm disposed in an appendix carry position, some tactical belts fail to allow for flexibility (e.g., “give”), which can cause the firearm to uncomfortably interact with the user. However, the inner belt portion **504** disclosed herein allows for the buckle **509** to maintain engagement while allowing for the first buckle portion **522-1** and the second buckle portion **522-2** to move apart slightly due to the inclusion of stiffening tongue **524** thereby allowing an amount of flexibility in the inner belt portion **504** that can secure the firearm without the uncomfortable interaction to the user mentioned above.

[0050] In addition, the stiffening tongue **524** provides rigidity to the inner belt portion **504**, thereby allowing for common firearm carry techniques, such as appendix carry, with minimal to no sag, while maintaining comfort for the user, as discussed in more detail below.

[0051] The stiffening tongue **524** can be formed using a range of composite materials that can contour to the body shape of a user of the self-aligning tactical belt over time. This allows for improved comfort in comparison to previous approaches while providing a zero to minimal sag of the inner belt portion **504** that can be used on its own for everyday conceal carry applications. That is, the composite materials used for the stiffening tongue **524** and/or the inner belt portion **504** can be selected from groups of composite materials that allow the inner belt portion **504** and/or the stiffening tongue **524** to gradually achieve a shape that contours to the body shape and/or size of the user thereby improving the comfort of the self-aligning tactical belt while also improving the strength, durability, and sag of the inner belt portion **504**, as well as the self-aligning tactical belt.

[0052] FIG. 6 illustrates an example of a method **630** for fabricating the self-aligning tactical belt (e.g., the self-aligning tactical belt **100** of FIG. 1). It should be noted that certain steps within the method **630** may be optional, and the steps shown in the method **630** are merely examples for illustration, and certain other steps may be included or excluded as desired. Further, while a particular order of the steps is shown, this ordering is merely illustrative, and any suitable arrangement of the steps may be utilized without departing from the scope of the embodiments herein. More-

over, various fabrication and/or manufacturing techniques can be utilized without departing from the scope of the disclosure.

[0053] At operation **631**, the method **630** can include forming an inner belt portion having an inner surface and an outer surface extending between a first end of the inner belt portion and a second end of the inner belt portion. The inner belt portion can be analogous to the inner belt portion **104/204/304/504** of FIG. 1, FIG. 2B, FIG. 3, and/or FIG. 5, herein.

[0054] At operation **632**, the method **630** can include forming a first magnetic component disposed between the inner surface and the outer surface of the inner belt portion on a posterior location of the inner belt portion. The first magnetic component disposed between the inner surface and the outer surface of the inner belt portion on the posterior location of the inner belt portion can be analogous to the first magnetic component **110-1/210-1/310-1/510-1** of FIG. 1, FIG. 2B, FIG. 3, and/or FIG. 5, herein.

[0055] At operation **633**, the method **630** can include forming a second magnetic component disposed between the inner surface and the outer surface of the inner belt portion on the posterior location of the inner belt portion. The second magnetic component disposed between the inner surface and the outer surface of the inner belt portion on the posterior location of the inner belt portion can be analogous to the second magnetic component **110-2/210-2/310-2/510-2** of FIG. 1, FIG. 2B, FIG. 3, and/or FIG. 5, herein.

[0056] At operation **634**, the method **630** can include forming an outer belt portion having an inner surface and an outer surface extending between a first end of the outer belt portion and a second end of the outer belt portion. The outer belt portion can be analogous to the outer belt portion **102/202/302/402** of FIG. 1, FIG. 2A, FIG. 4A, and/or FIG. 4B, herein.

[0057] At operation **635**, the method **630** can include forming a third magnetic component disposed between the inner surface and the outer surface of the outer belt portion on a posterior location of the outer belt portion. The third magnetic component disposed between the inner surface and the outer surface of the outer belt portion on the posterior location of the outer belt portion can be analogous to the first magnetic component **110-1/210-1/310-1/510-1** of FIG. 1, FIG. 2B, FIG. 3, and/or FIG. 5, herein. At operation **636**, the method **630** can include forming a fourth magnetic component disposed between the inner surface and the outer surface of the outer belt portion on the posterior location of the outer belt portion. The fourth magnetic component disposed between the inner surface and the outer surface of the outer belt portion on the posterior location of the outer belt portion can be analogous to the second magnetic component **110-2/210-2/310-2/510-2** of FIG. 1, FIG. 2B, FIG. 3, and/or FIG. 5, herein.

[0058] As described in more detail above, the first magnetic component disposed between the inner surface and the outer surface of the inner belt portion is couplable to the third magnetic component disposed between the inner surface and the outer surface of the outer belt portion to facilitate self-alignment of the inner belt portion and the outer belt portion and the second magnetic component disposed between the inner surface and the outer surface of the inner belt portion is couplable to the fourth magnetic component disposed between the inner surface and the outer

surface of the outer belt portion to facilitate self-alignment of the inner belt portion and the outer belt portion.

[0059] The mechanisms described herein, therefore, introduce a self-aligning tactical belt and/or an apparatus, system, and/or method for making the same in a manner that achieves a rapidly installable, repeatable, and symmetric fitting for a tactical belt having two separable components (e.g., an inner belt portion and an outer belt portion). Further, the described configurations minimize the amount of time and instances that a wearer must attempt to couple the outer belt portion to the inner belt portion to achieve a perfect coupling where the user can immediately engage the buckle and utilize the self-aligning tactical belt. That is, as a result of the magnetic components discussed herein, the outer belt portion may be automatically aligned with the inner belt portion for a rapid and foolproof coupling of the outer belt portion to the inner belt portion thereby saving time and frustration for the user.

[0060] According to the embodiments herein, an illustrative tactical belt (e.g., the self-aligning tactical belt described herein) may comprise: an inner belt portion having, on a posterior area of the inner belt portion, a first inner belt magnetic component and a second inner belt magnetic component coupled thereto and an outer belt portion having, on a posterior area of the outer belt portion, a first outer belt magnetic component and a second outer belt magnetic component coupled thereto where the first inner belt magnetic component and the second inner belt magnetic component are configured to couple to the first outer belt magnetic component and the second outer belt magnetic component, respectively, to facilitate alignment of the inner belt portion and the outer belt portion.

[0061] In one embodiment, the outer belt portion includes a skeletonized reinforcement band coupled to the outer belt portion that extends beyond a first end of the outer belt portion such that, when the outer belt portion is coupled to the inner belt portion, the skeletonized reinforcement band extends beyond a second end of the outer belt portion to form a contiguous ring. In such embodiments, the first outer belt magnetic component and the second outer belt magnetic component are spaced at a distance that corresponds to a spacing associated with the skeletonized reinforcement band. In addition, when a webbing is affixed to the skeletonized reinforcement band, the webbing can extend beyond a second end of the outer belt portion to form a contiguous ring of webbing. Further, in some embodiments, the webbing can be provided such that the first outer belt magnetic component and the second outer belt magnetic component are spaced at a distance that corresponds to a spacing associated with the webbing.

[0062] In one embodiment, the inner belt portion includes a stiffening tongue on an anterior area of the inner belt portion. In such embodiments, the inner belt portion further includes a buckle mechanism having a first dimension on a first end of the buckle mechanism and a second dimension on a second end of the buckle mechanism, the first end of buckle mechanism and the second end of the buckle mechanism coupled to the stiffening tongue and the buckle mechanism is configured to expand to expose a portion of the stiffening tongue in response to a movement of a user of the tactical belt.

[0063] In one embodiment, the self-aligning tactical belt includes a first hook-and-loop fastener on at least a portion of an outer surface of the inner belt portion and a second

hook-and-loop fastener on at least a portion of an outer surface of the outer belt portion. In such embodiments, the first hook-and-loop fastener is couplable to the second hook-and-loop fastener to connect the inner belt portion to the outer belt portion.

[0064] In one embodiment, the first outer belt magnetic component and the second outer belt magnetic component have an opposite magnetic polarity than the first inner belt magnetic component and the second inner belt magnetic component. Embodiments are not so limited, however, and in some embodiments, the first outer belt magnetic component can have a first polarity (e.g., a north polarity) and the first inner belt magnetic component can have an opposite polarity (e.g., a south polarity), while the second outer belt magnetic component can have a second polarity (e.g., a south polarity) and the second inner belt magnetic component can have an opposite polarity (e.g., a north polarity). It is noted that providing magnetic components that have opposite polarities can enhance the self-aligning features of the tactical belt in comparison to approaches in which a metal component is provided to interface with the magnetic components.

[0065] According to the embodiments herein, an illustrative tactical belt (e.g., the self-aligning tactical belt described herein) may comprise: an inner belt portion having an inner surface and an outer surface extending between a first end of the inner belt portion and a second end of the inner belt portion. The inner belt portion includes: a first magnetic component disposed between the inner surface and the outer surface of the inner belt portion; and a second magnetic component disposed between the inner surface and the outer surface of the inner belt portion. The first magnetic component and the second magnetic component are disposed on a posterior location of the inner belt portion.

[0066] In such embodiments, the illustrative tactical belt (e.g., the self-aligning tactical belt described herein) may further comprise: an outer belt portion having an inner surface and an outer surface extending between a first end of the outer belt portion and a second end of the outer belt portion. The outer belt portion includes a third magnetic component disposed between the inner surface and the outer surface of the outer belt portion; and a fourth magnetic component disposed between the inner surface and the outer surface of the outer belt portion. In these embodiments, the third magnetic component and the fourth magnetic component are disposed on a posterior location of the outer belt portion.

[0067] In one embodiment, the inner belt portion includes a first portion of a first buckle coupled to the first end of the inner belt portion and a second portion of the first buckle coupled to the second end of the inner belt portion. In such embodiments, the outer belt portion includes a first portion of a second buckle coupled to the first end of the outer belt portion and a second portion of the second buckle coupled to the second end of the outer belt portion.

[0068] In one embodiment, the first magnetic component is located in a position on the inner belt portion that aligns with a position of the third magnetic component when the inner belt portion and the outer belt portion are coupled together and the second magnetic component is located in a position on the inner belt portion that aligns with a position of the fourth magnetic component when the inner belt portion and the outer belt portion are coupled together. In such embodiments, the first magnetic, the second magnetic com-

ponent, the third magnetic component, and the fourth magnetic component provide for the inner belt portion and the outer belt portion to be automatically aligned when the inner belt portion and the outer belt portion are coupled together.

[0069] In one embodiment, the outer belt portion includes a skeletonized reinforcement band coupled to the outer belt portion that extends beyond the first end of the outer belt portion such that, when the outer belt portion is fastened, the skeletonized reinforcement band extends through the second end of the outer belt portion to form a contiguous ring. In such embodiments, the first magnetic component and the second magnetic component are spaced at a distance that corresponds to a spacing (e.g., a spacing associated with the edges and voids of the skeletonized reinforcement band) associated with the skeletonized reinforcement band when the inner belt portion is coupled to the outer belt portion and/or the third magnetic component and the fourth magnetic component are spaced at a distance that corresponds to a spacing associated with the skeletonized reinforcement band when the inner belt portion is coupled to the outer belt portion.

[0070] As mentioned above, when a webbing is affixed to the skeletonized reinforcement band, the webbing can extend beyond a second end of the outer belt portion to form a contiguous ring of webbing. Further, in some embodiments, the webbing can be provided such that the first outer belt magnetic component and the second outer belt magnetic component are spaced at a distance that corresponds to a spacing associated with the webbing.

[0071] In such embodiments, the at least one row of webbing comprises a Modular Lightweight Load-carrying Equipment (MOLLE) webbing. Further, in such embodiments, the first magnetic component and the second magnetic component are spaced at a distance that corresponds to a spacing associated with webs of the at least one row of webbing when the inner belt portion is coupled to the outer belt portion and/or the third magnetic component and the fourth magnetic component are spaced at a distance that corresponds to a spacing associated with webs of the at least one row of webbing.

[0072] In one embodiment, the self-aligning tactical belt includes a first hook-and-loop fastener on at least a portion of the outer surface of the inner belt portion and a second hook-and-loop fastener on at least a portion of the outer surface of the outer belt portion. In such embodiments, the first hook-and-loop fastener is couplable to the second hook-and-loop fastener to connect the inner belt portion to the outer belt portion.

[0073] In one embodiment, the first magnetic component has a first polarity and the third magnetic component has a second polarity and the second magnetic component has the second polarity and the fourth magnetic component has the first polarity. For example, the first polarity can be a north polarity and the second polarity can be a south polarity, or vice versa.

[0074] In one embodiment, the inner belt portion further includes a buckle mechanism having a first dimension on a first end of the buckle mechanism and a second dimension on a second end of the buckle mechanism and the buckle mechanism is configured to expand in response to a movement of a user of the tactical belt.

[0075] According to the embodiments herein, an illustrative tactical belt (e.g., the self-aligning tactical belt described herein) may be fabricated by: forming an inner belt portion

having an inner surface and an outer surface extending between a first end of the inner belt portion and a second end of the inner belt portion; forming a first magnetic component disposed between the inner surface and the outer surface of the inner belt portion on a posterior location of the inner belt portion; forming a second magnetic component disposed between the inner surface and the outer surface of the inner belt portion on the posterior location of the inner belt portion; forming an outer belt portion having an inner surface and an outer surface extending between a first end of the outer belt portion and a second end of the outer belt portion; forming a third magnetic component disposed between the inner surface and the outer surface of the outer belt portion on a posterior location of the outer belt portion; and forming a fourth magnetic component disposed between the inner surface and the outer surface of the outer belt portion on the posterior location of the outer belt portion.

[0076] In one embodiment, the tactical belt may be fabricated such that the first magnetic component is couplable to the third magnetic component to facilitate self-alignment of the inner belt portion and the outer belt portion, and the second magnetic component is couplable to the fourth magnetic component to facilitate self-alignment of the inner belt portion and the outer belt portion.

[0077] The foregoing description has been directed to specific embodiments. It will be apparent, however, that other variations and modifications may be made to the described embodiments, with the attainment of some or all of their advantages. The construction and arrangement of the self-aligning tactical belt as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.).

[0078] For example, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied.

[0079] Accordingly, all such modifications are intended to be included within the scope of the present disclosure. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

[0080] Accordingly, this description is to be taken only by way of example and not to otherwise limit the scope of the embodiments herein. Therefore, it is the object of the appended claims to cover all such variations and modifications as come within the true spirit and scope of the embodiments herein.

What is claimed is:

1. A tactical belt, comprising:

an inner belt portion having, on a posterior area of the inner belt portion, a first inner belt magnetic component and a second inner belt magnetic component coupled thereto; and

an outer belt portion having, on a posterior area of the outer belt portion, a first outer belt magnetic component and a second outer belt magnetic component coupled thereto, wherein:

the first inner belt magnetic component and the second inner belt magnetic component are configured to couple to the first outer belt magnetic component and

the second outer belt magnetic component, respectively, to facilitate alignment of the inner belt portion and the outer belt portion.

2. The tactical belt of claim 1, wherein the outer belt portion includes a skeletonized reinforcement band coupled to the outer belt portion that extends beyond a first end of the outer belt portion such that, when the outer belt portion is coupled to the inner belt portion, the skeletonized reinforcement band extends beyond a second end of the outer belt portion to form a contiguous ring.

3. The tactical belt of claim 2, wherein the first outer belt magnetic component and the second outer belt magnetic component are spaced at a distance that corresponds to a spacing associated with areas of the skeletonized reinforcement band.

4. The tactical belt of claim 1, wherein the inner belt portion includes a stiffening tongue on an anterior area of the inner belt portion.

5. The tactical belt of claim 4, wherein:

the inner belt portion further includes a buckle mechanism having a first dimension on a first end of the buckle mechanism and a second dimension on a second end of the buckle mechanism, the first end of buckle mechanism and the second end of the buckle mechanism coupled to the stiffening tongue, and

the buckle mechanism is configured to expand to expose a portion of the stiffening tongue in response to a movement of a user of the tactical belt.

6. The tactical belt of claim 1, further comprising:

a first hook-and-loop fastener on at least a portion of an outer surface of the inner belt portion; and

a second hook-and-loop fastener on at least a portion of an outer surface of the outer belt portion, wherein:

the first hook-and-loop fastener is couplable to the second hook-and-loop fastener to connect the inner belt portion to the outer belt portion.

7. The tactical belt of claim 1, wherein the first outer belt magnetic component and the second outer belt magnetic component have an opposite magnetic polarity than the first inner belt magnetic component and the second inner belt magnetic component.

8. A tactical belt, comprising:

an inner belt portion having an inner surface and an outer surface extending between a first end of the inner belt portion and a second end of the inner belt portion, wherein the inner belt portion includes:

a first magnetic component disposed between the inner surface and the outer surface of the inner belt portion; and

a second magnetic component disposed between the inner surface and the outer surface of the inner belt portion, wherein:

the first magnetic component and the second magnetic component are disposed on a posterior location of the inner belt portion; and

an outer belt portion having an inner surface and an outer surface extending between a first end of the outer belt portion and a second end of the outer belt portion, wherein the outer belt portion includes:

a third magnetic component disposed between the inner surface and the outer surface of the outer belt portion; and

a fourth magnetic component disposed between the inner surface and the outer surface of the outer belt portion, wherein:

the third magnetic component and the fourth magnetic component are disposed on a posterior location of the outer belt portion, and wherein:

the first magnetic component is couplable to the third magnetic component, and

the second magnetic component is couplable to the fourth magnetic component.

9. The tactical belt of claim 8, wherein the inner belt portion includes:

a first portion of a first buckle coupled to the first end of the inner belt portion; and

a second portion of the first buckle coupled to the second end of the inner belt portion, and wherein the outer belt portion includes:

a first portion of a second buckle coupled to the first end of the outer belt portion; and

a second portion of the second buckle coupled to the second end of the outer belt portion.

10. The tactical belt of claim 8, wherein:

the first magnetic component is located in a position on the inner belt portion that aligns with a position of the third magnetic component when the inner belt portion and the outer belt portion are coupled together, and

the second magnetic component is located in a position on the inner belt portion that aligns with a position of the fourth magnetic component when the inner belt portion and the outer belt portion are coupled together.

11. The tactical belt of claim 10, wherein the first magnetic component, the second magnetic component, the third magnetic component, and the fourth magnetic component provide for the inner belt portion and the outer belt portion to be automatically aligned when the inner belt portion and the outer belt portion are coupled together.

12. The tactical belt of claim 8, wherein the outer belt portion includes a skeletonized reinforcement band coupled to the outer belt portion that extends beyond the first end of the outer belt portion such that, when the outer belt portion is fastened, at least a portion of the skeletonized reinforcement band extends through the second end of the outer belt portion to form a contiguous ring.

13. The tactical belt of claim 12, wherein the skeletonized reinforcement band includes spacing associated with a Modular Lightweight Load-carrying Equipment (MOLLE) webbing.

14. The tactical belt of claim 12, wherein the first magnetic component and the second magnetic component are spaced at a distance that corresponds to a spacing associated with the skeletonized reinforcement band when the inner belt portion is coupled to the outer belt portion.

15. The tactical belt of claim 12, wherein the third magnetic component and the fourth magnetic component are spaced at a distance that corresponds to a spacing associated with the skeletonized reinforcement band.

16. The tactical belt of claim 8, further comprising:

a first hook-and-loop fastener on at least a portion of the outer surface of the inner belt portion; and

a second hook-and-loop fastener on at least a portion of the outer surface of the outer belt portion, wherein:

the first hook-and-loop fastener is couplable to the second hook-and-loop fastener to connect the inner belt portion to the outer belt portion.

17. The tactical belt of claim **8**, wherein:

the first magnetic component has a first polarity and the third magnetic component has a second polarity, and the second magnetic component has the second polarity and the fourth magnetic component has the first polarity.

18. The tactical belt of claim **8**, wherein:

the inner belt portion further includes a buckle mechanism having a first dimension on a first end of the buckle mechanism and a second dimension on a second end of the buckle mechanism, and

the buckle mechanism is configured to expand in response to a movement of a user of the tactical belt.

19. A method for fabricating a tactical belt, comprising:

forming an inner belt portion having an inner surface and an outer surface extending between a first end of the inner belt portion and a second end of the inner belt portion;

forming a first magnetic component disposed between the inner surface and the outer surface of the inner belt portion on a posterior location of the inner belt portion;

forming a second magnetic component disposed between the inner surface and the outer surface of the inner belt portion on the posterior location of the inner belt portion;

forming an outer belt portion having an inner surface and an outer surface extending between a first end of the outer belt portion and a second end of the outer belt portion;

forming a third magnetic component disposed between the inner surface and the outer surface of the outer belt portion on a posterior location of the outer belt portion; and

forming a fourth magnetic component disposed between the inner surface and the outer surface of the outer belt portion on the posterior location of the outer belt portion.

20. The method of claim **19**, wherein:

the first magnetic component is couplable to the third magnetic component to facilitate self-alignment of the inner belt portion and the outer belt portion, and the second magnetic component is couplable to the fourth magnetic component to facilitate self-alignment of the inner belt portion and the outer belt portion.

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